NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE NAME

BIOLOGY CLASS

REGISTRATION NUMBER

BIOLOGY

9648/03

Paper 3

15 September 2015

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A
Answer all questions.
Write your answers in this booklet using dark blue or black pen.

Section B
Write your answers in this booklet using dark blue or black pen.
For Free Response Question, write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
1	/ 10
2	/ 10
3	/ 10
4	/ 10
5	/ 12
6	/ 20
TOTAL	**/ 72**

This document consists of **14** printed pages.

[Turn over

Section A

Answer all questions.

1 Fig. 1.1 shows the plasmid pACE15 which was isolated from bacteria and genetically engineered to contain the two selectable markers *tet*^{R} (confers cell resistance to antibiotic tetracyclin) and *lacZ* (encodes enzyme ß–galactosidase).

pACE15 is currently being used in biotechnology industries for the mass production of human growth hormone (hGH).In recombinant plasmids, the hGH gene is present within the Lac Z gene. Recombinant plasmids are subsequently transformed into *E. coli* cells. The bacterial cells from the transformation process are grown on an agar plate containing tetracycline and X-Gal.

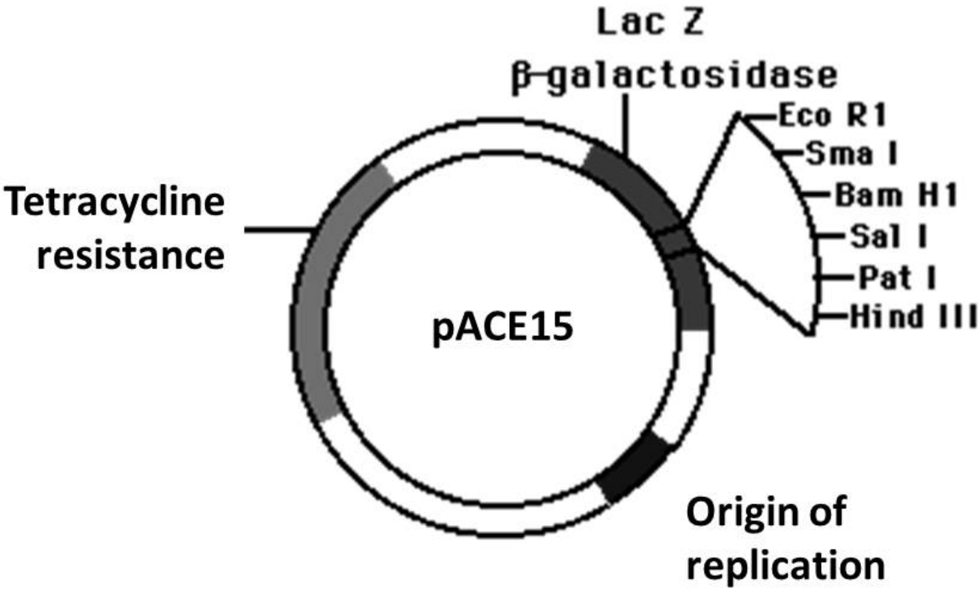

Fig. 1.1

(a) With reference to Figure 1.1, explain the feature of plasmid pACE15 that allows it to be used for gene cloning involving any species of organism. [2]

Table 1.1 shows two restriction enzymes that were considered for the process of inserting the hGH gene into the plasmid to form recombinant DNA.

Table 1.1

Restriction enzyme	**Restriction site**
EcoRI	**5’ GAATTC 3’** **3’ CTTAAG 5’**
SmaI	**5’ CCCGGG 3’** **3’ GGGCCC 5’**

(b) With reference to Table 1.1, state and explain which restriction enzyme would be more suitable for use in genetic engineering. [2]

(c) With reference to Fig. 1.1, explain how transformed cells containing recombinant plasmids are differentiated from transformed cells containing re-annealed non-recombinant plasmids. [2]

(d) Describe how hGH gene in recombinant plasmids can be induced to undergo expression in *E.coli*. [1]

(e) Suggest the advantage of using yeast cells instead of bacteria for the production of hGH protein. [2]

[Total: 9]

2 Phenylketonuria (PKU) is an autosomal recessive inherited genetic disorder of metabolism where mutations in the PAH gene cause low levels of an enzyme called phenylalanine hydroxylase, that affects metabolism of phenylalanine. Without dietary treatment, phenylalanine can build up to harmful levels in the body, causing mental retardation and other serious problems.

A DNA segment, PK-21 found within 5 Centimorgans of the PAH gene has been identified. This segment exhibits restriction fragment length polymorphism (RFLP) that can be used as a marker for the genetic screening of PKU.

Fig. 2.1 shows the RFLPs of PK-21.

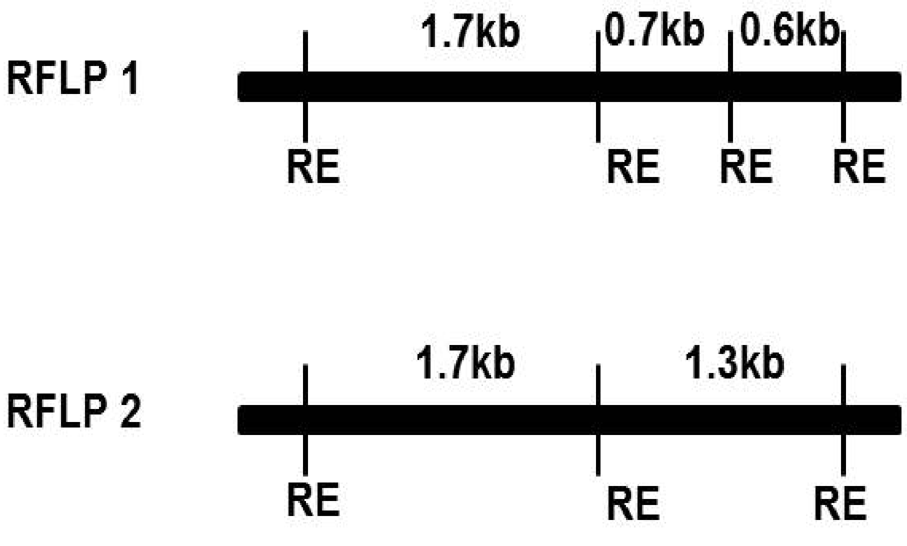

Fig. 2.1

Fig 2.2 shows the pedigree for the family and Fig 2.3 shows the genetic screening results for a family including that of the foetus the mother is carrying.

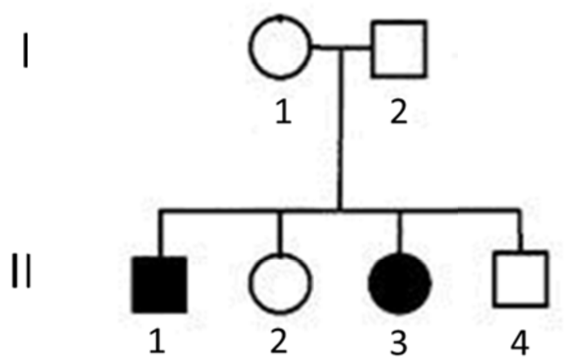

Fig. 2.2

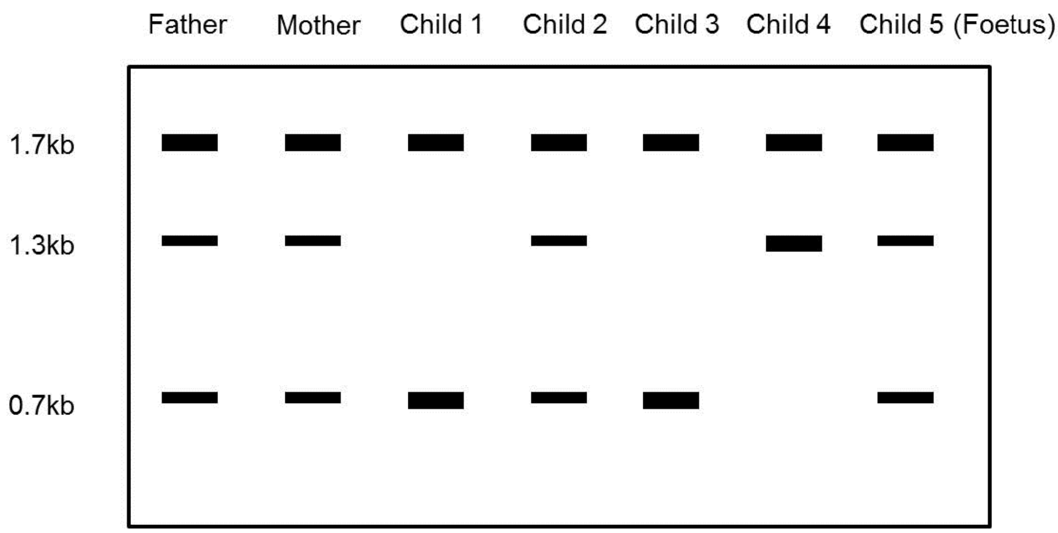

Fig. 2.3

(a) With reference to Fig. 2.1, explain one reason why PK-21 can be used as a genetic marker for the detection of the mutant PAH allele. [2]

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(b) The 2 RFLPs are detected using a probe P. With reference to Fig. 2.1 and 2.3, indicate on Fig 2.1, the position that probe P will bind. [1]

(c) Explain how gel electrophoresis is used to analyse and visualize DNA. [3]

(d) With reference to Fig. 2.2 and 2.3, explain which RFLP the mutant PAH allele is linked to. [2]

(e) The newborn (Child 5) is diagnosed with PKU upon birth. Suggest a reason to explain the banding pattern observed for Child 5 in the genetic screening results. [1]

[Total: 9]

3 Duchenne muscular dystrophy (DMD) is a genetic disease cuased by the absence of the protein dystrophin in muscle fibres. In the absence of dystrophin, muscle fibres gradually die.

A potential gene therapy for DMD involves injecting mucles with a viral vector carrying recombinant DNA (rDNA) for part of the normal allele for dystrophin.

(a) Outline the formation of recombinant DNA. [3]

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(b) Mice with the symptoms of DMD were given this gene therapy shortly after birth. Each mouse was injected with the viral vector in a muscle of one hind limb. The corresponding muscle of the other hind limb was injected with a buffer solution to provide a control. The nuclei of muscle fibres that do not produce dystrophin move from the edge of the fibre to the centre. The fibres eventually die. The percentage of muscle fibres with centrally placed nuclei was measured in fibres from treated and control muscles at different times after injection. The results are shown in the figure below.

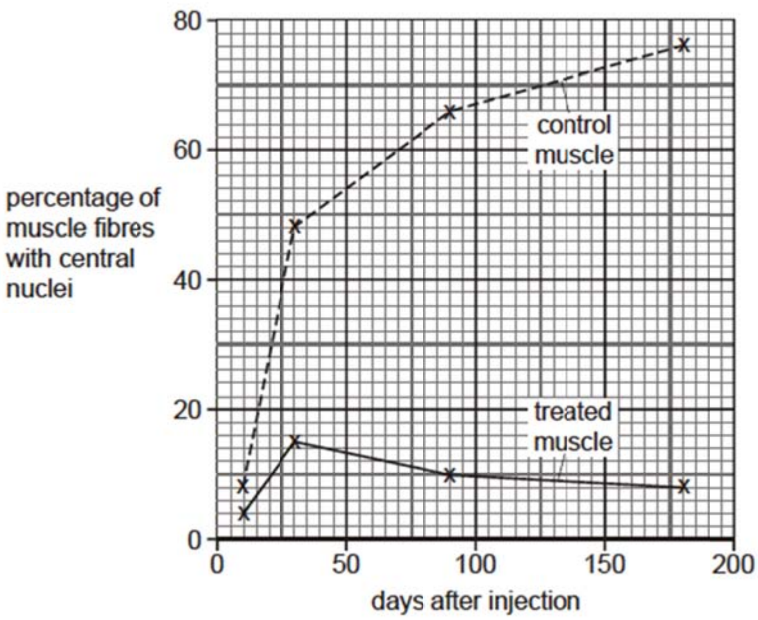

Fig. 3.1

(i) Identify the type of gene therapy treatment stated above [1]

(ii) Using the information above, describe the results of the experiment. [3]

(iii) Explain why it is easier to perform gene therapy when the normal allele is the dominant allele of the gene concerned. [2]

(iv) Genetic screening is available for families with a history of DMD. Discuss the advantages and disadvantages of genetic screening. [4]

[Total: 13]

Name: ________________________ Class: 2bi2___ / 2IPbi2___

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Section B

Answer **all** the questions in this section.

4 Experimental results below show the vulnerability of using various Orchid plant parts in carrying out micropropagation, at various times of the year.

Table 4.1

explant	time of year	number of explants	number of cultured explants without any microbe contamination	Percentage of cultured explants without microbe contamination/ %
leaf disc	May	153	12	**8**
	September	322	16	**5**
	February	332	30	**9**
stem tissue	May	194	116	**60**
	September	191	122	**64**
	February	211	156	**74**

(a) With reference to **Table 4.1**, compare the effect of the different methods of taking tissue samples and the time of year on microbe contamination of the cultured explants. [3]

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(b) Suggest why it is crucial for the removal of pathogens from the explants. [2]

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(c) Explain the advantages of using cloned plants in the farming of orchids. [3]

(d) State one concern of farming genetically modified herbicide resistant corn and suggest a way of overcoming it. [2]

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[Total: 10]

5 Planning Question

Write your answers to this question on the separate answer paper provided.

You are required to plan an experiment to determine the effects of the growth regulator gibberellin (GA) on the germination of maize grains.

You are required to use the following information to assist in your experiment plan:

- Maize needs to be soaked in water for 24 hours to stimulate germination
- Maize starts to germinate 48 – 72 hours after soaking
- GA promotes germination by activating a gene that causes the synthesis of the enzyme amylase
- The range of concentrations at which GA is found in plants is between 1 – 10 μmol dm-3
- Amylase is released by the aleurone layer into the endosperm of the maize grain to hydrolyse the starch reserves
- The activity of amylase can be estimated by cutting the maize grains lengthways into two halves and placing the cut sides onto agar containing Starch in Petri dishes
- After incubation at a constant temperature, iodine solution is used to test the agar for the presence of starch

Your planning must be based on the assumption that you have also been provided with the following materials which you must use:

- 3 mmol dm-3 GA solution

In addition, you may include in your plan the usage of any equipment that can be found in a standard school laboratory.

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure.
- a description of the method used, including the scientific reasoning behind the method
- an explanation of the dependent, independent variables and controlled variables
- how you will record your results and ensure that they are as accurate & reliable as possible.
- proposed layout of results tables and graph with clear headings and labels
- recommended safety measure.
- the correct use of technical and scientific terms.

[Total: 12]

Free Response Question

Write your answers to this question on the separate answer paper provided.
Begin each section of the question on a new piece of answer paper.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b), etc., as indicated in the question.

6 **(a)** Discuss the factors which prevent gene therapy from becoming an effective treatment. [8]

(b) Using blood stem cells as example, describe the unique features and functions of stem cells. Explain the differences between stem cells and cancer cells. [12]

[Total: 20]

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